

MathFoundations230:

**Multiset arithmetic
via trees**

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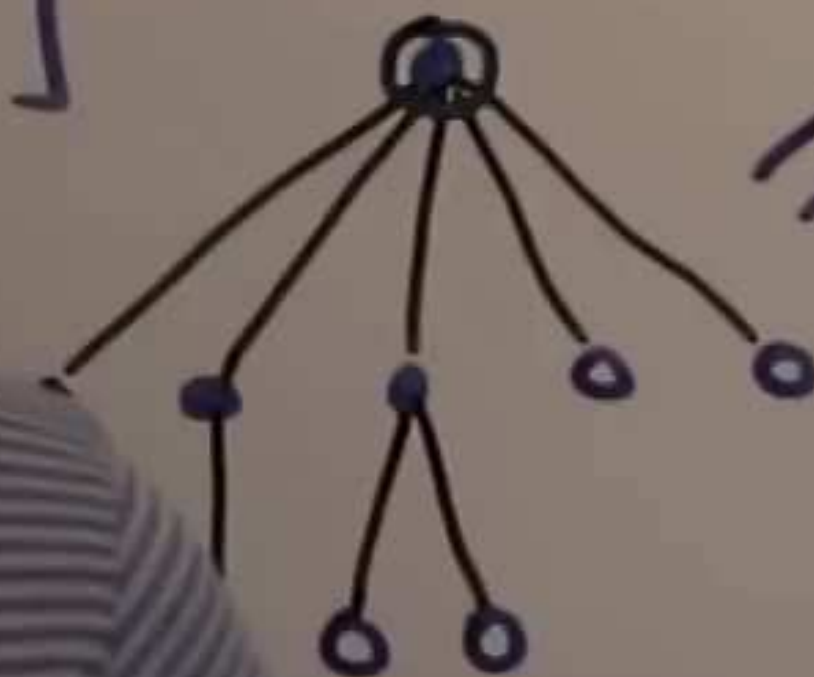
MF230: Multiset arithmetic via trees

$$A \equiv \begin{bmatrix} [] & [] & [] & [] \\ [0] & [0] & [0] & [0] \\ [1] & [1] \end{bmatrix}$$

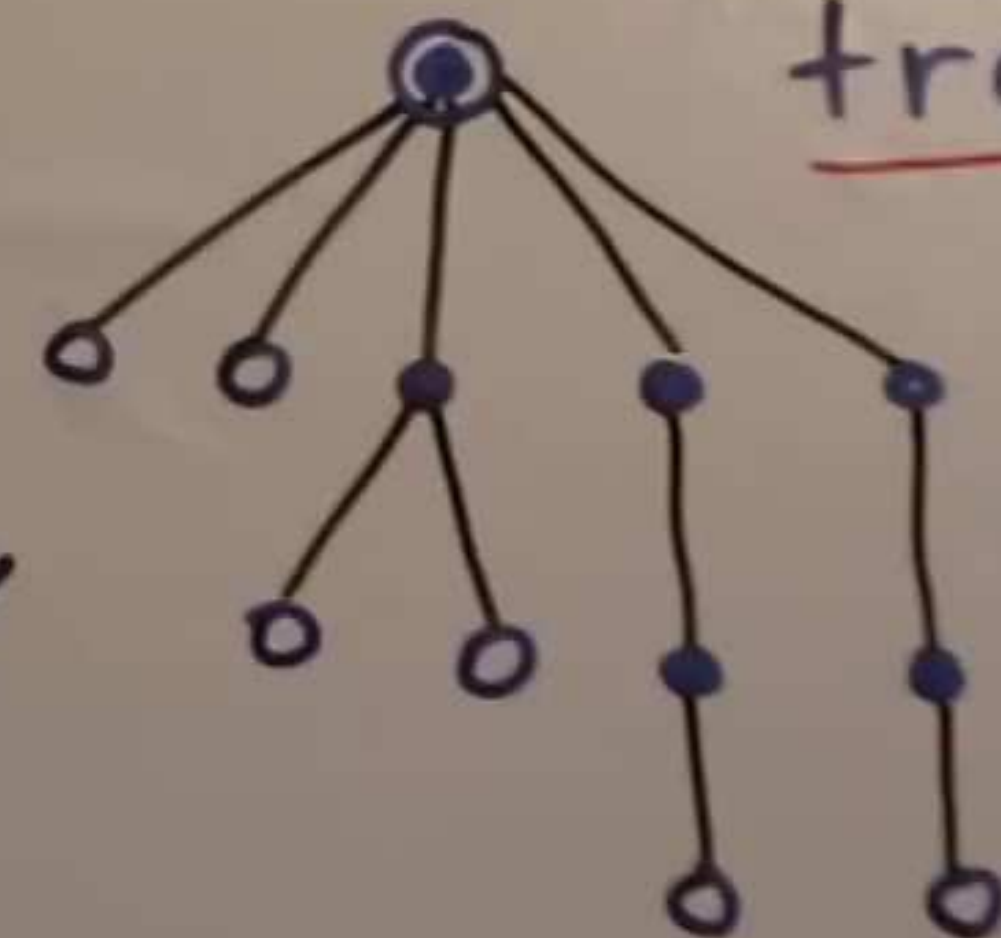
$$= \begin{bmatrix} [0] & [0] & [0] \end{bmatrix}$$

$$= \begin{bmatrix} [1] & [1] \end{bmatrix}$$

rooted/
rooted
tree



=

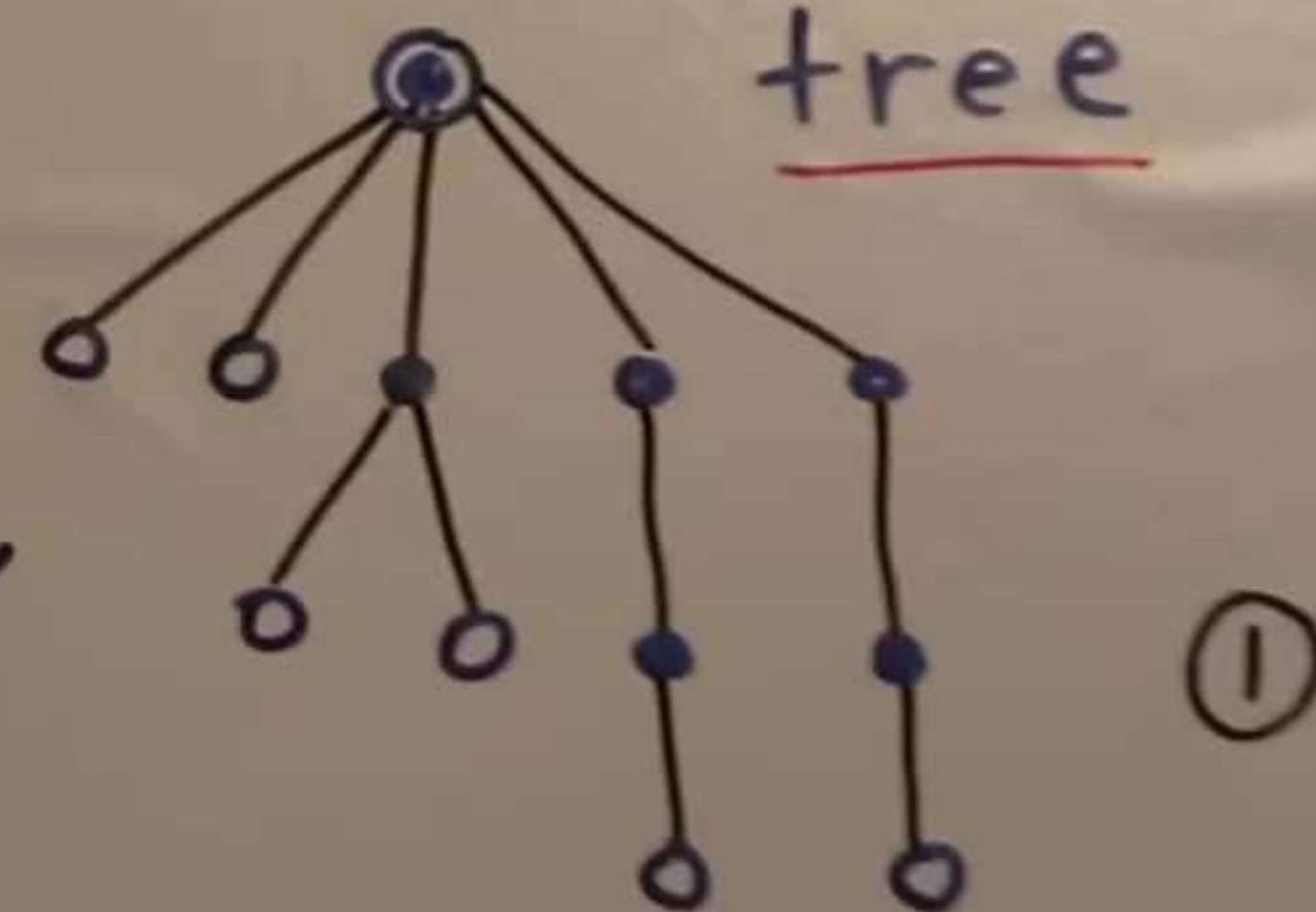


①

MF230: Multiset arithmetic via trees

$$A = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

rooted/
rooted
tree

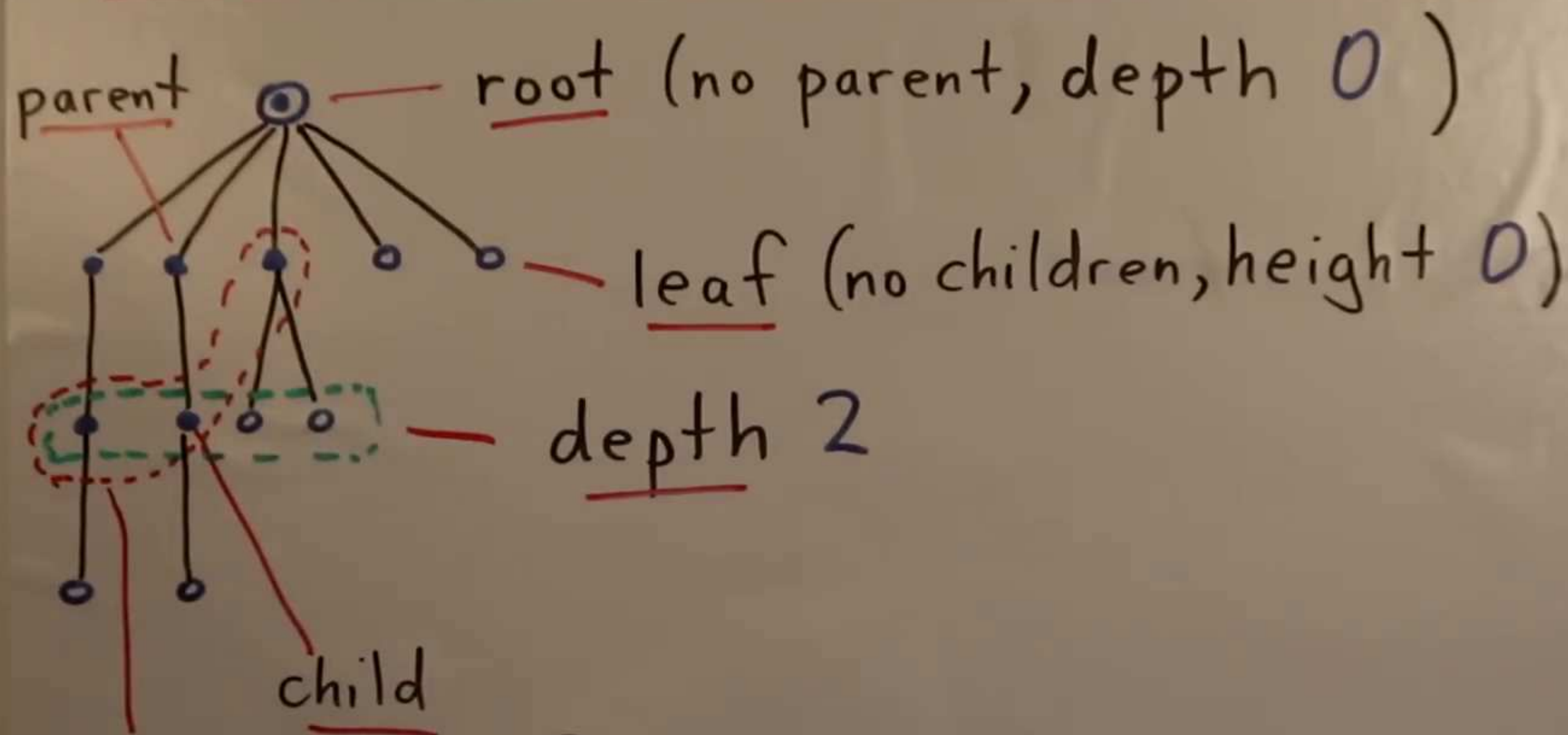


Notation for rooted/roofed trees
root (no parent, depth 0)
leaf (no children, height 0)
depth 2



$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Notation for rooted/roofed trees



②

$$\text{height 1 } A \equiv \left[\left[\right] \left[\right] \left[\left[\right] \left[\right] \right] \left[\left[\left[\right] \right] \left[\left[\right] \right] \right] \right]$$

Notation for rooted/roofed trees

parent

parent → root (no parent, depth 0)

- leaf (no children, height 0)

Multiplicity convention (left subindices)

$A = [[[3 [[3 [3]] [[[3]]]]]]]$

$= [[[3 [[3 [3]]]]]] + [[[3 [[3]]]]] + 2 \times [[[[3]]]]$

$= [[[3 [[3 [3]]]]]] + [[[3 [[3]]]]] + 2 [[[[3]]]]$

③

Multiplicity convention (left)

$A = [[[3] [3] [[[3] [3]]]]] + [[[3] [3]]] + 2 \times [[[3] [3]]]$



Multiplicity convention (left sub indices)

$$A = \begin{bmatrix} [] & [] & [[] []] & [[[]]] & [[[]]] \end{bmatrix}$$

$$[\quad] + [[\quad] [\quad]] + 2 \times [[[\quad]]]$$

③

Counting with trees

Counting with tree

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

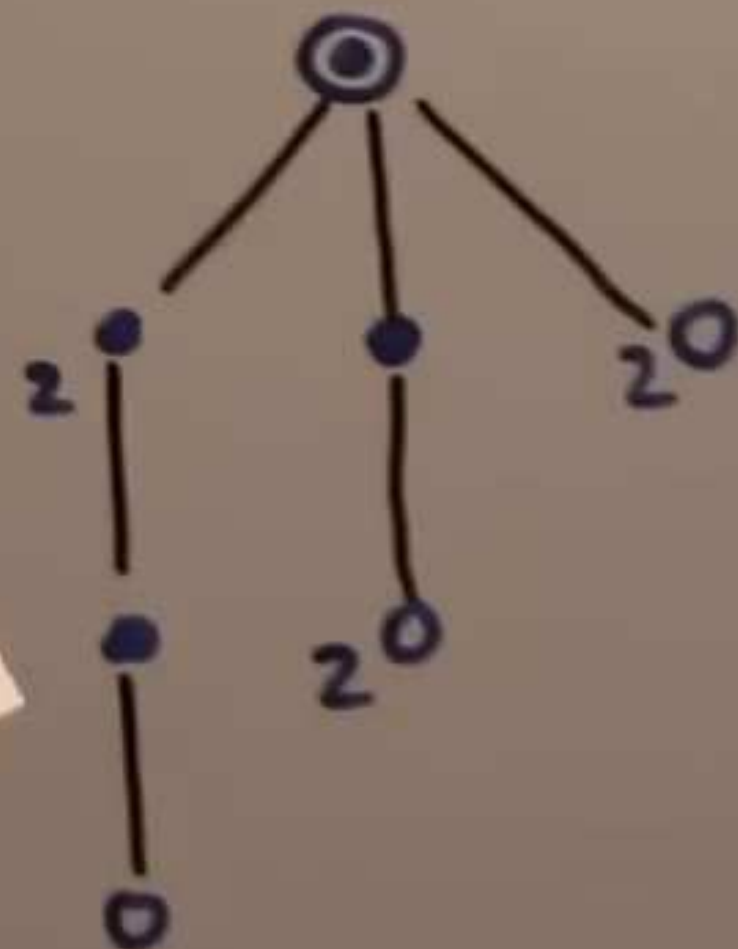
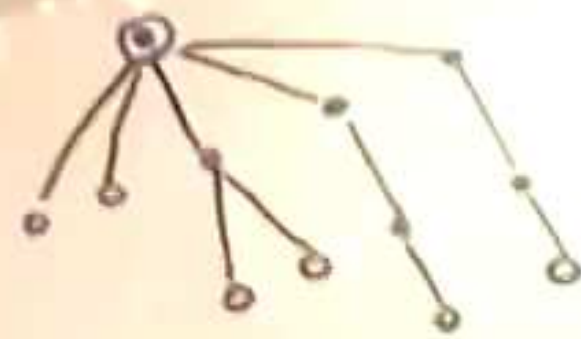
$$\chi(a \in A) = [1 \ 1 \ 1 \ 1 \ 1] = 5$$

$$[a \in A] = [0] + [0] + [0]$$

$$|A| = 2 + 2\alpha_0 + \alpha_0^2$$

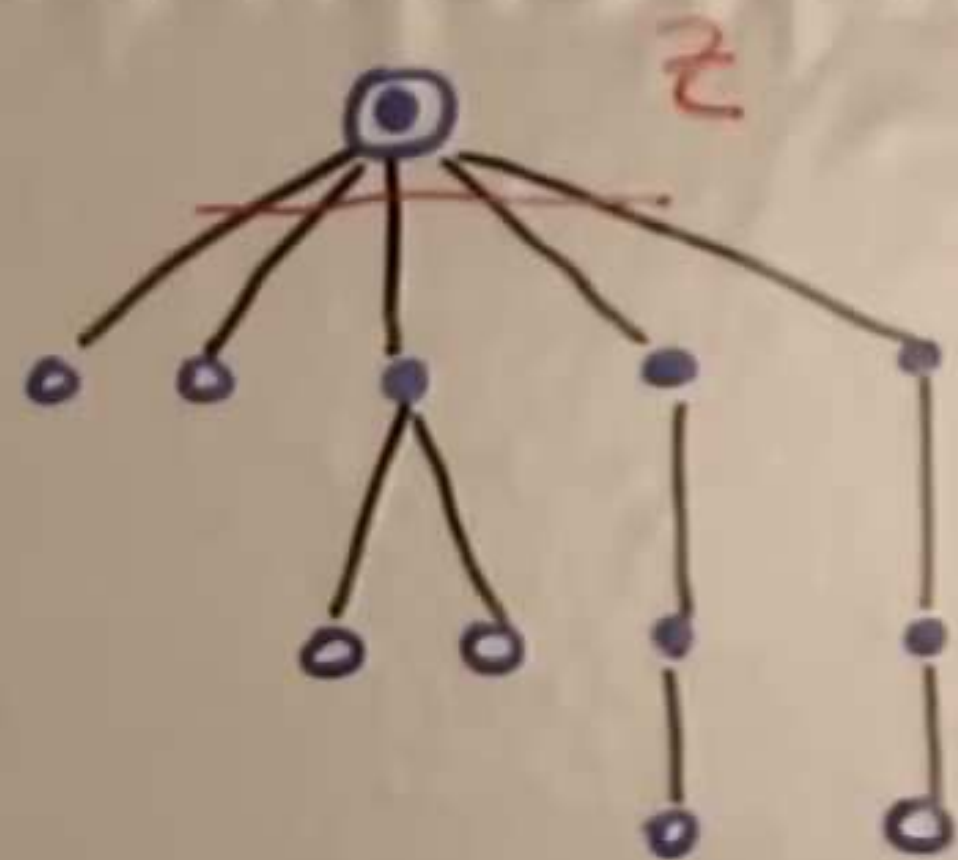
$\mathbb{A}^1 = \mathbb{A}$

Counting = Truncation



Counting with trees

$$A = \left[\begin{array}{c} [] \\ [] \\ [] \\ [] \\ [] \end{array} \right] A \equiv$$



$$\text{Then } Z(A) = 0 = []$$

$$Z(A) = [Z(a) : a \in A] = \left[\begin{array}{c} [] \\ [] \\ [] \\ [] \\ [] \end{array} \right] = [00000] = 5$$

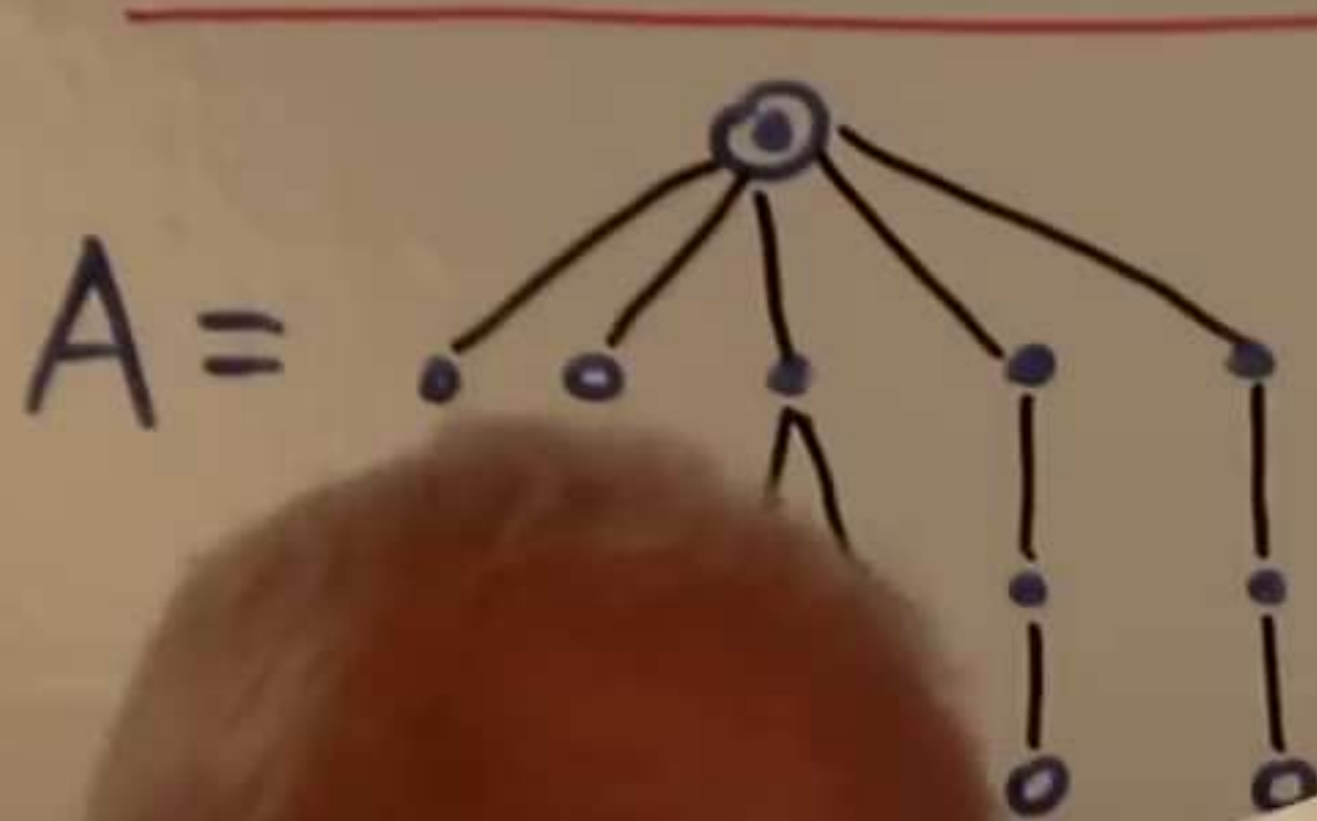
$$P(A) = [N(a) : a \in A] = \left[\begin{array}{c} [] \\ [] \\ [0 \ 0] \\ [0] \\ [0] \end{array} \right] \quad (4)$$
$$= 2 + 2\alpha_0 + \alpha_0^2$$

$$M(A) = [P(a) : a \in A] = A$$

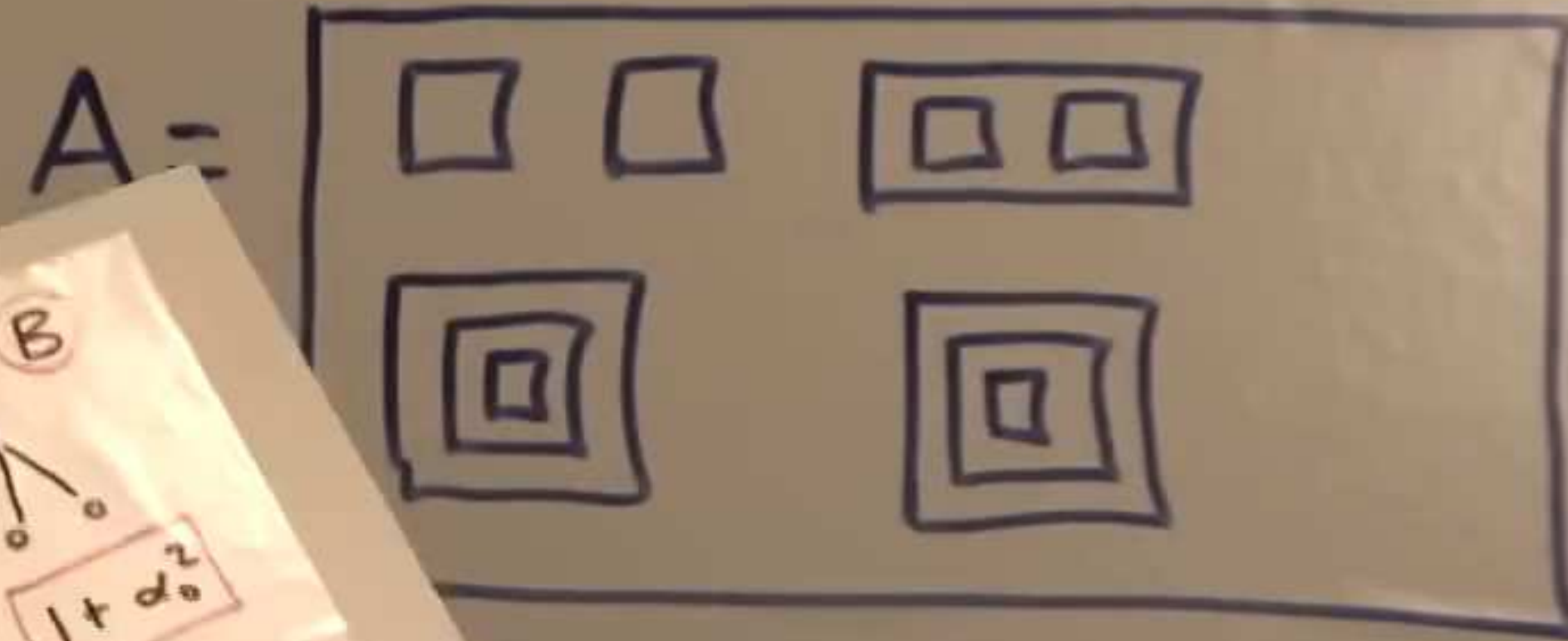
Counting = Truncation

Counting with trees

Closed boxes/nodes

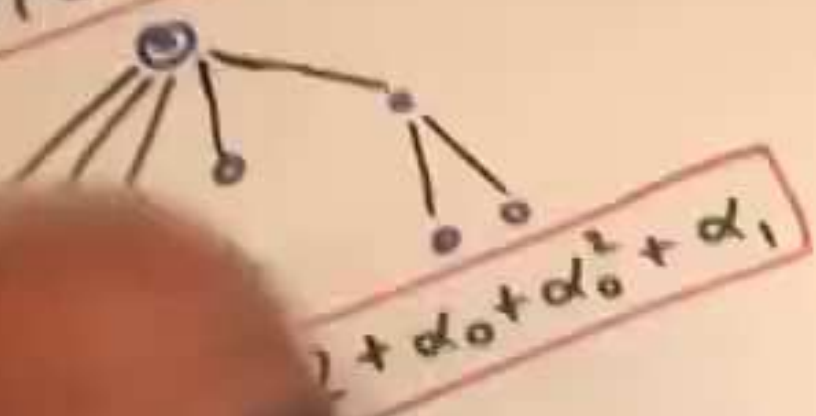


$$A = \begin{bmatrix} 0 & 0 & [0 & 0] & [1] & [1] \end{bmatrix}$$

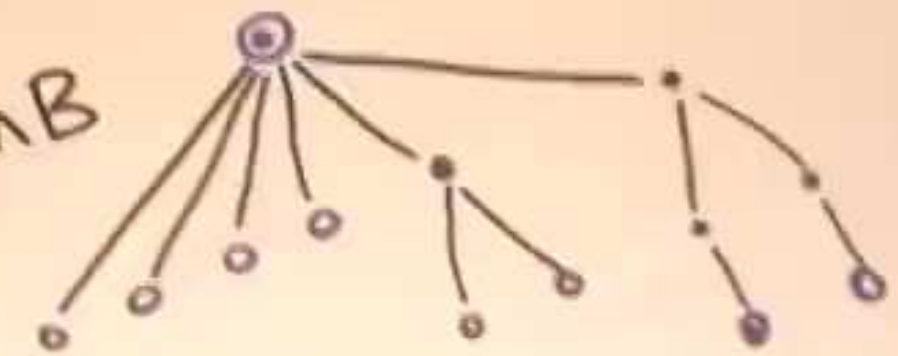


Arithmetic with trees

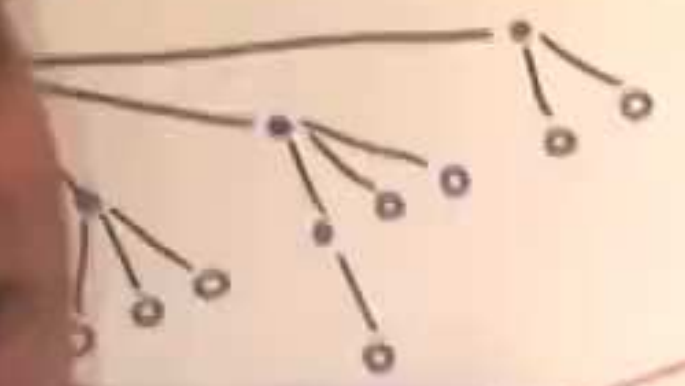
A+B



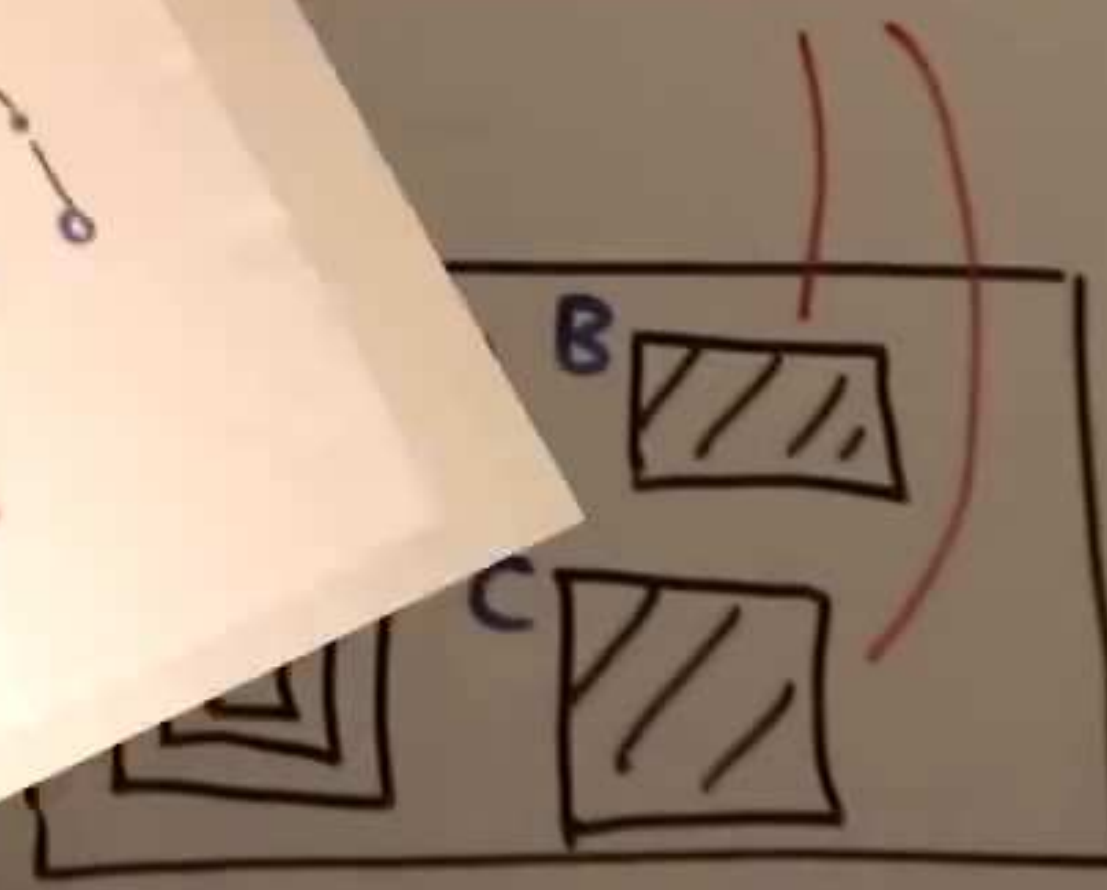
A \wedge B



A \times



used boxes

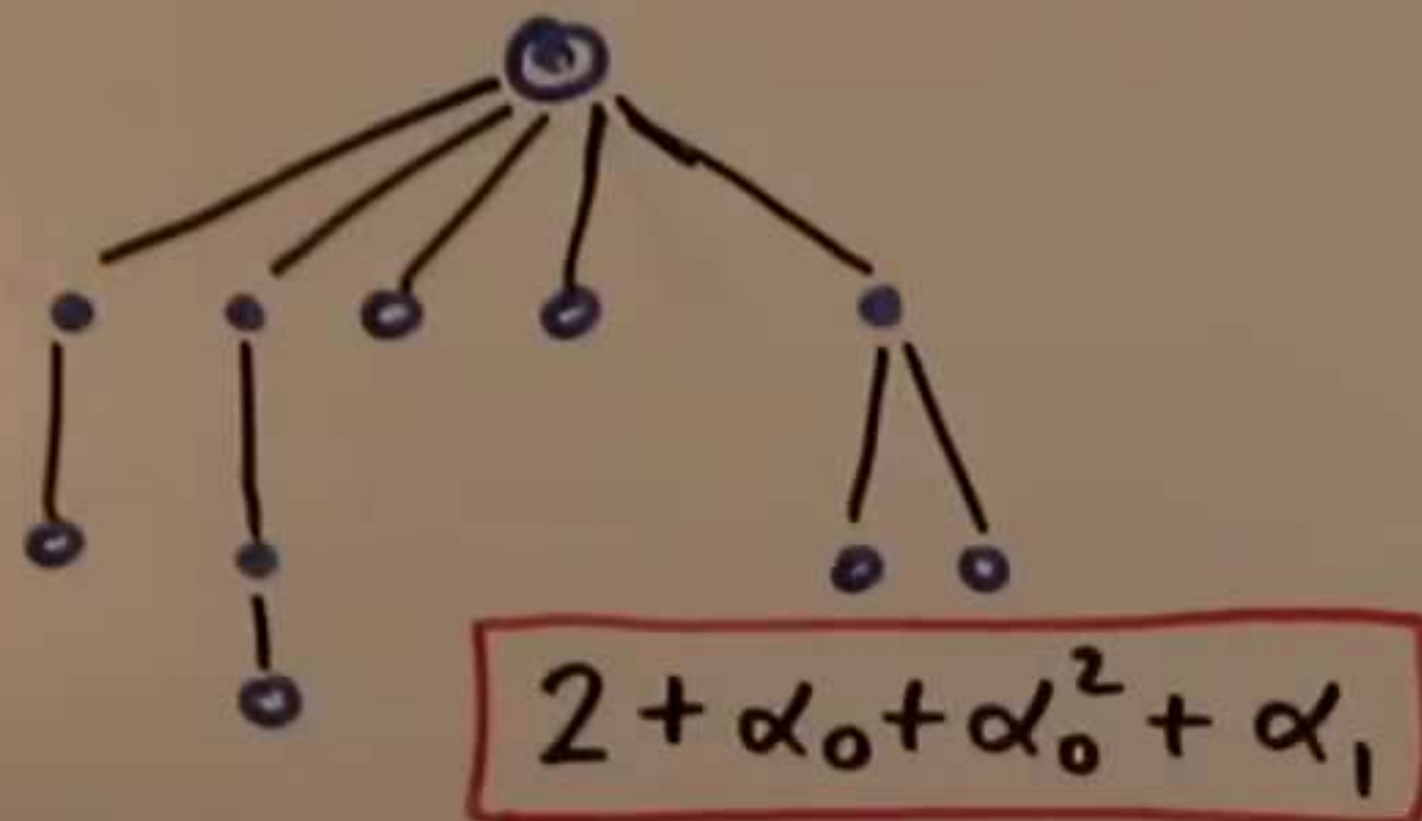


$$B = \begin{bmatrix} \square & \square \end{bmatrix}$$

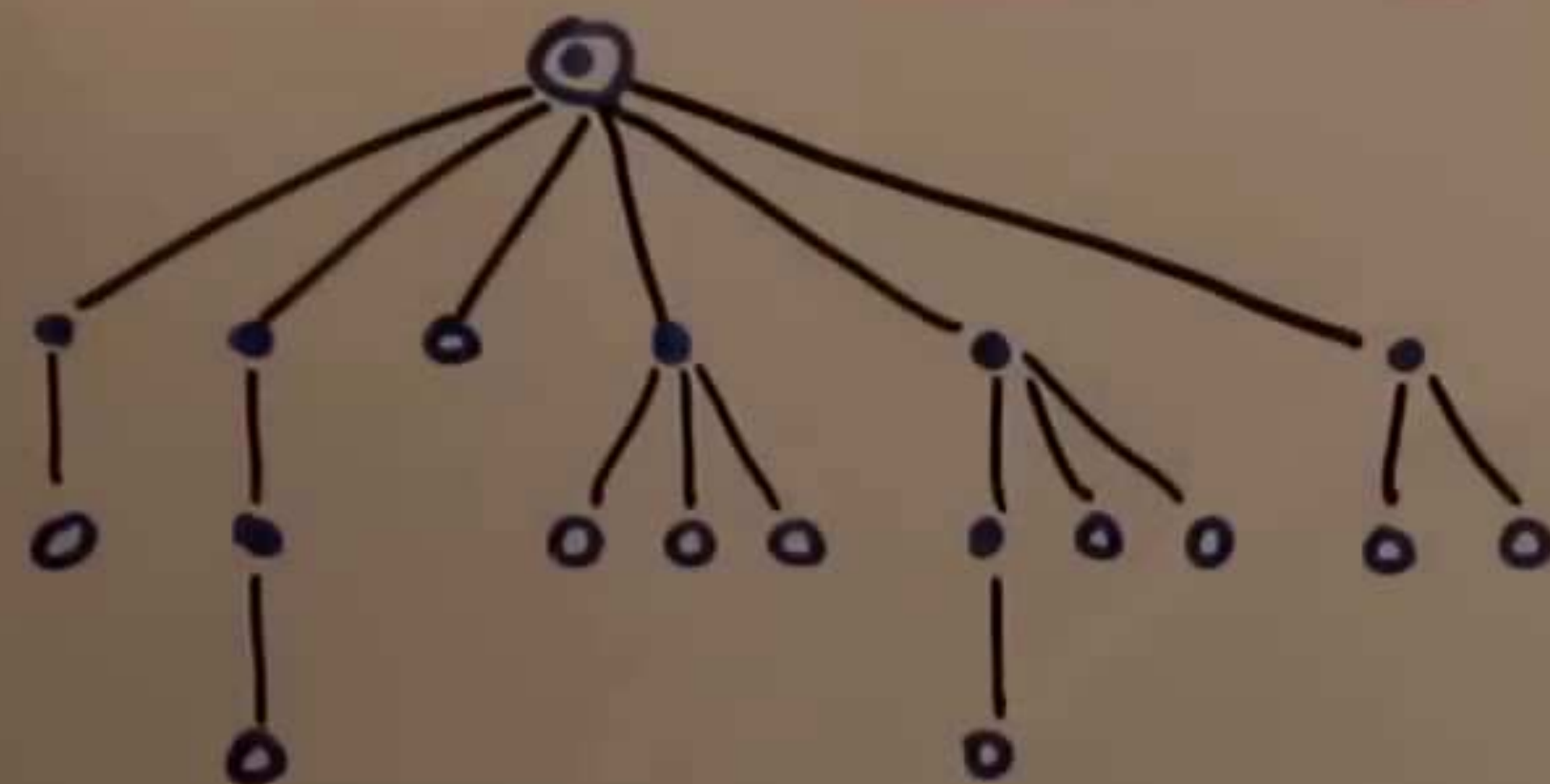
$$C = \begin{bmatrix} \square \end{bmatrix}$$

Arithmetic with trees

$A+B$



$A \times B$

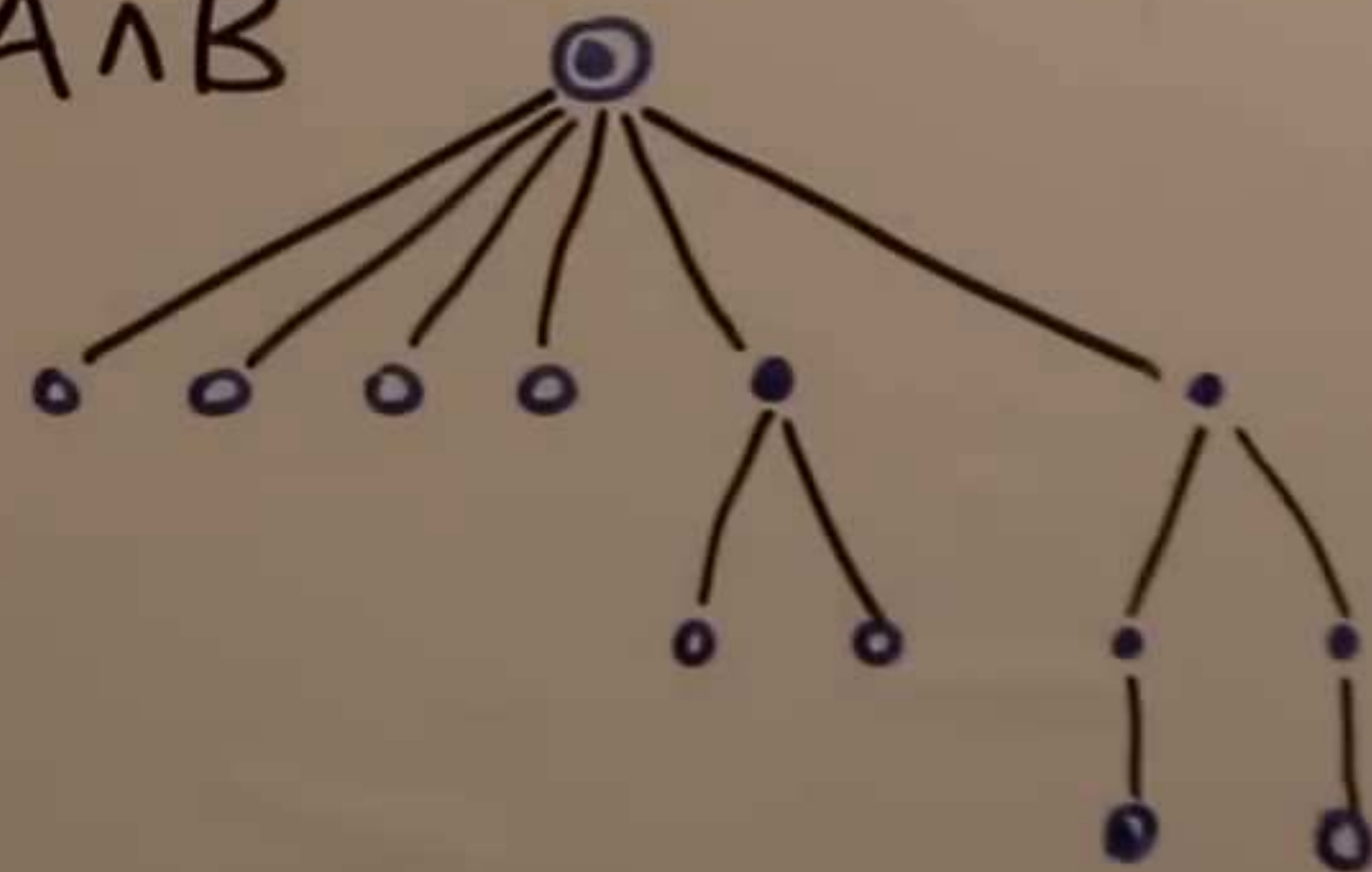


A



B

$A \wedge B$



$\textcircled{6}$

Next in the series:

MathFoundations231:

**Negative numbers from
anti multisets**